Course Transfer Request

Short Summary

Request	SYSU Course	CMU Course	Grade (Out of 100)
1	Advanced Algorithms and Programming Techniques	15-750 Algorithms in the Real World	88
2	Stochastic Processes	15-857 Analytical Performance Modeling and Design of Computer Systems	85
3	Machine Learning	10-701 Introduction to Machine Learning	85
4	Pattern Recognition & Computational Visual Media	11-777 Multimodal Machine Learning	88 & 90
5	Advanced Computer Networks	15641 Networking and the Internet	92

Notes:

- All SYSU courses were taught at the graduate level, 3 credits (54 contact hours over 18 weeks; 3 contact hours per week).
- Each course was graded on a 100-point scale, with 60 as the minimum passing score.
- The courses of Request 4 jointly correspond to a single CMU course (11-777), based on content coverage.

The following 5 sections present each request. Each section begins with key information (e.g., file list, webpage) and is followed by three parts: (1) Course Information, (2) Course Description, and (3) Topic Mapping and Content Equivalence Comparison.

Request 1

SYSU Advanced Algorithms and Programming Techniques ↔
CMU 15-750 Algorithms in the Real World

SYSU Course Assignment & Slides (attached files):

- **Assignment**
- **Slices**
    - 1. Introduction
    - 2. Linear Programming
    - 3. Network Flow Algorithms
    - 4. Algorithm Analysis and Computational Complexity
    - 5. NP-Completeness
    - 6. Approximation Algorithms
    - 7. Randomized Algorithms
    - 8. Strategies for Dealing with Intractable Problems

- 9. Review

CMU Course Reference Webpage: https://www.cs.cmu.edu/~15750/

1. Course Information

Category	SYSU Course	CMU Course
Course Title	Advanced Algorithms and Programming Techniques	Algorithms in the Real World
Institution	Sun Yat-sen University (SYSU), School of Computer Science and Engineering	Carnegie Mellon University (CMU), School of Computer Science
Course Code	-	15-750
Instructor	Prof. Zhiming Dai daizhim@mail.sysu.edu.cn	Prof. Jason Li
Credit Hours	3 credits (54 contact hours)	12 units
Textbook	Algorithm Design and Analysis, second edition, by Wanling Qu, Tian Liu, Lian'ang Zhang, and Hanpin Wang.	No mandatory textbook for the course
Assignment	Programing project: Apply Max-flow–Min-cut theorem to perform image segmentation. · Implement the algorithm. · Provide flowchart, mathematical derivation, and output examples. · Explain computational complexity.	Six bi-weekly homeworks (mix of analytical and implementation problems; individual and collaborative parts)
Exam	Final Exam	Midterm and Final Exams
Grading	10% participation, 40% homework, 50% final exam	32% for the midterm, 32% for the final exam, 32% for the homeworks, and 4% for attendance/class participation in lectures/Piazza/office hours.
Course Webpage	-	https://www.cs.cmu.edu/~15750/

2. Course Description Comparison

Aspect	SYSU: Advanced Algorithms and Programming Techniques	CMU: 15-750 Algorithms in the Real World

Course Focus	Provides an in-depth study of algorithm design and analysis techniques at the graduate level. Strengthens theoretical understanding and applied skills through both mathematical analysis and a programming project. Topic: Divide-and-Conquer, Dynamic Programming, Greedy Algorithms, Branch-and-Bound, Linear Programming, Network Flows, NP-Completeness, Approximation and Randomized Algorithms	Covers a broad set of algorithmic tools and their applications. Aims to teach students to (a) recognize appropriate methods, (b) apply them effectively, and (c) reason about correctness and performance in real-world contexts. Topics: Greedy, Divide-and-Conquer, Dynamic Programming, Hashing and Randomization, String Algorithms, Streaming (Big Data) Algorithms, High-Dimensional Data (JL Transform, Nearest Neighbor), Flows and Cuts, Linear and Convex Programming, Compression, Error Correction, Gradient Descent and Multiplicative Weights, Online Decision Making
Learning Objectives	1. Design efficient algorithms using classical paradigms (DP, greedy, flow). 2. Analyze algorithm efficiency and complexity. 3. Understand NP-completeness and develop approximation or randomized solutions for hard problems. 4. Implement and evaluate algorithms on real data.	-

3. Topic Mapping and Content Equivalence

CMU 15-750 Lecture	SYSU Coverage
L1 – Introduction, Triangle Counting, Strassen	Partially Covered. Complexity analysis, asymptotics, and recurrences appear in Slides 1 & 4. These provide the analytical foundation for divide-and-conquer algorithms such as Strassen. Triangle counting relates conceptually to graph-theoretic ideas discussed in Slides 4 & 5 (reductions, combinatorial reasoning).
L2 – MST & Amortized Analysis	Partially Covered. MST-related concepts appear indirectly in approximation and graph-based NP-complete problems (Slides 5 & 6). General amortized-style reasoning is supported by complexity analysis foundations in Slide 4.

L3 – Union-Find	Conceptually Covered. While SYSU does not explicitly teach DSU, the algorithmic role of Union-Find in graph algorithms is implicitly supported through problem reductions and NP-hard graph problems in Slide 5, and complexity analysis in Slide 4.
L4 – Shortest Paths	Mostly Covered. Bellman-Ford and Floyd-Warshall appear explicitly in Slide 3 (Network Flow Algorithms). The shortest-path paradigm is covered conceptually, though Dijkstra's algorithm is not explicitly included.
L5 – Dynamic Programming	Fully Covered. SYSU includes DP paradigms, examples, and applications in Slide 1.
L6–7 – Hashing, Load Balancing, Chernoff Bounds	Partially Covered. SYSU teaches Markov/Chernoff bounds, expectation/variance, and probabilistic inequalities in Slide 7, providing the mathematical backbone for hashing analysis and load-balancing concentration arguments. Hashing itself is not explicitly covered.
L8 – Streaming Algorithms	Conceptually Related. While streaming and sketching algorithms do not appear directly, SYSU covers complexity models, pseudo-polynomial behaviors, and NP-hardness limits in Slides 4 & 5, which are foundational to lower-bound arguments in streaming.
L9 – Dimensionality Reduction (JL Transform)	Conceptually Related. SYSU's randomized-algorithm tools (Slide 7) and linear-algebraic LP foundations (Slide 2) form the mathematical basis for JL matrices and probabilistic embeddings, though JL is not explicitly taught.
L10–11 – Tries & Suffix Structures	Algorithmic Paradigm Overlap. SYSU's dynamic programming, greedy approaches, and algorithmic complexity (Slides 1, 4) cover the high-level paradigms used in modern string algorithms, although trie, suffix tree, and suffix array structures themselves do not appear in any slide.
L12 – SVD	Indirectly Supported. Linear-algebraic reasoning from LP geometry (Slide 2) gives conceptual prerequisites for SVD-based algorithmic thinking, though SVD is not explicitly included in the curriculum.
L13 – Max-Flow, Matchings	Fully Covered. SYSU covers max-flow/min-cut, Ford–Fulkerson, Edmonds–Karp, Dinic, and bipartite matching extensively in Slide 3.
L14 – Min-Cost Flow	Fully Covered. SYSU provides complete treatment of min-cost flow and negative-cycle-based algorithms in Slide 3.

L15 – Linear Programming	Fully Covered. SYSU has extensive LP content including standard form, simplex, duality, ILP, and branch-and-bound in Slide 2.
L16–18 – Multiplicative Weights, Zero-Sum Games	Partially Covered (Conceptually). SYSU covers LP relaxations, approximation frameworks, and error-bound reasoning in Slides 6 & 8, which conceptually overlap with the optimization and regret-based ideas underlying multiplicative weights, though MWU itself is not taught.
L18b – Gradient Descent, Convex Optimization	Conceptually Covered. Optimization thinking, LP geometric interpretation, and feasible-region reasoning in Slide 2 provide foundational ideas for convex optimization and gradient descent, though no explicit convex-optimization algorithms appear.

All major design paradigms and optimization techniques (divide-and-conquer, DP, flows, LP, randomization) appear in both courses.

Request 2

SYSU Stochastic Processes ↔
CMU 15-857 Analytical Performance Modeling and Design of Computer Systems

SYSU Course Assignment & Slides (attached files)

- **Textbooks**
- **Assignments:**
  - Homework-1
  - Homework-2
  - Homework-3
  - Homework-4
  - Homework-5
  - Homework-6
  - Homework-7
  - Homework-8
  - Homework-9
  - Homework-10
  - Homework-11: Programming assignment
  - Homework-12
- **Slides:**
  - SP01-Introduction
  - SP02-Typical Probability Models
  - SP03-Bounds and Limits
  - SP04-Arrival Processes
  - SP05-Properties of the Poisson Process
  - SP06-Finite-state Markov Chains (DTMC)
  - SP07-Stochastic Matrices (Steady-State)
  - SP08-Continuous Time Markov Chain (CTMC)

- SP09-Queuing Theory
- SP10-Markov Decision Processes (MDP)
- SP11-Finite Horizon MDP
- SP12-Infinite Horizon MDP & Project Assignment
- SP13-Approximate Dynamic Programming & Reinforcement Learning
- SP14-Renewal Processes
- SP15-Stopping Times

CMU Course Reference Webpag: https://www.cs.cmu.edu/~harchol/Perfclass/class.html

1. Course Information

Category	SYSU Course	CMU Course
Course Title	Stochastic Processes	Analytical Performance Modeling and Design of Computer Systems
Institution	Sun Yat-sen University (SYSU), School of Computer Science and Engineering	Carnegie Mellon University, School of Computer Science
Course Code	-	15-857
Instructor	Prof. Xiao Ma, Prof. Jie Gong maxiao@mail.sysu.edu.cn gongj26@mail.sysu.edu.cn	Prof. Mor Harchol-Balter
Credit Hours	3 credits (54 contact hours)	12 units
Textbook	1. Robert G. Gallager, Stochastic Processes: Theory for Applications, Cambridge University Press, 2013. 2. Michael Mitzenmacher and Eli Upfal, Probability and Computing: Randomization and Probabilistic Techniques in Algorithms and Data Analysis (second edition), Cambridge University Press, 2017. 3. Dimitri P. Bertsekas, Dynamic Programming and Optimal Control,Vol. I, 3rd Edition, Athena Scientific, 2005.	Performance Modeling and Design of Computer Systems https://www.cs.cmu.edu/~harchol/PerformanceModeling/book.html
Assignment	**11 Theoretical Quizzes**: covering topics from Poisson Processes, Markov Chain analysis, to MDP policy optimization. **1 Programming Project:** • Formulate a "batch manufacturing problem" as both discounted and average cost MDPs. • Implement core algorithms: Value Iteration and Policy Iteration. • Analyze and compare the optimal policies under different parameters.	• Weekly Homeworks • ONE mandatory grading meeting during semester. • ONE 30-minute individual or pair meeting with Mor where you present an example of queueing in your research or your life. Best to run this by a TA first! Super informal. No slides.

Exam	Final Exam	2 Midterm Exams
Grading	10% Attendance, 40% Coursework (11 Quizzes, 1 Final Programming Project), 50% Final Exam	• Weekly homework sets (40% of grade), • 2 midterms (25% each). • ONE mandatory grading meeting -- 5% . • ONE 30-minute individual or pair meeting. -- 5% .
Course Webpage	-	https://www.cs.cmu.edu/~harchol/Perfclass/class.html

2. Course Description Comparison

Aspect	**SYSU Stochastic Processes**	**CMU 15-857 Analytical Performance Modeling and Design of Computer Systems**
Course Focus	Provides a comprehensive introduction to probability models and stochastic processes for engineering applications. Topics include probability bounds and limits, Poisson and renewal processes, discrete and continuous-time Markov chains, queueing theory, and Markov decision processes. Emphasizes mathematical rigor and engineering applications (communication systems, operations research, reinforcement learning).	This class will introduce students to analytic stochastic modeling. Topics covered include: 1. Operational Laws: Little's Law, response-time law, asymptotic bounds, modification analysis, performance metrics. 2. Markov Chain Theory: both discrete-time and continuous-time, renewal theory, time-reversibility. 3. Poisson Process: memorylessness, Bernoulli splitting, uniformity, PASTA. 4. Queueing Theory: open systems, closed systems, M/M/1, M/M/k, M/M/k/k, M/G/1 full analysis, M/G/k, G/G/1, transform analysis (Laplace and z-transforms). 5. Queueing Theory: Networks of Queues. Jackson and class-Jackson networks. 6. Simulations: time averages versus ensemble averages, generating random variables for simulation, Inspection Paradox. 7. Modeling Empirical Workloads: heavy-tailed property, Pareto distributions, heavy-tailed distributions, understanding variability and tail behavior. 8. Management of Server Farms: capacity provisioning, dynamic power management, routing policies. 9. Analysis of Scheduling: FCFS, non-preemptive priorities, preemptive priorities, PS, LCFS, FB, SJF, PSJF, SRPT, plus the latest scheduling research: SOAP, Gittins Index, Whittle Index. 10. Applications to Today's Datacenters: Scheduling for multiserver system, resource

		allocation for multi-dimensional jobs, c-mu rule for maximizing value, parallel jobs with different speedup functions.
Learning Objectives	1. Build a rigorous foundation of probability and random process theory. 2. Model random systems using Poisson and Markov processes. 3. Analyze steady-state and transient behavior of stochastic systems. 4. Apply Markov decision process and dynamic programming for stochastic control problems. 5. Use simulation for queueing and MDP systems.	-

3. Topic Mapping and Content Equivalence

15-857 Topic	**SYSU Correspondence**
Operational Laws: Little's Law, response-time law, asymptotic bounds, modification analysis, performance metrics	Partially Covered — SYSU presents Little's Law and core performance metrics (L, W, ρ) in queueing models (SP08, SP09). Concepts like response-time law or asymptotic bounds are Conceptually Related through renewal and limiting arguments (SP14, SP15). Modification analysis does not explicitly appear.
Markov Chain Theory: DTMC, CTMC, renewal theory, time-reversibility	Substantially Covered — DTMC (classification, steady state) (SP06, SP07), CTMC with generator (Q) and Kolmogorov equations (SP08), and renewal theory (SP14) are all thoroughly treated. Time-reversibility is Conceptually Related to steady-state balance but not explicitly formalized.
Poisson Process: memorylessness, Bernoulli splitting, uniformity, PASTA	Substantially Covered — Independence, stationary increments, exponential interarrival times, memorylessness, Poisson superposition and splitting, and conditional uniformity of arrival times are directly presented (SP04, SP05). PASTA is Conceptually Related via Poisson arrivals and renewal reward arguments, though not explicitly named.

Queueing Theory: open/closed systems, M/M/1, M/M/k, M/M/k/k, M/G/1, M/G/k, G/G/1, transform analysis	Partially Covered — SYSU fully develops M/M/1, M/M/s, M/M/s(k), M/M/1(k) using CTMC/birth–death models (SP08, SP09). Renewal and stopping-time techniques give Indirect Support for M/G/1 and G/G/1 reasoning (SP14, SP15). Closed networks and transform-based analysis are Conceptually Related but not directly developed.
Queueing Networks: Jackson networks, class-Jackson networks	Conceptually Related — The building blocks (renewal, Poisson, CTMC, M/M/s queues) appear across SP06–SP09, but queueing networks themselves are not explicitly treated.
Simulations: time vs ensemble averages, RV generation, Inspection Paradox	Partially Covered / Indirectly Supported — SYSU offers substantial material on Monte Carlo simulation and random-variable generation (inverse CDF, uniform-based methods) (SP02). Time-average vs ensemble-average ideas are Conceptually Related via renewal-reward theorems (SP14), though not formalized. Inspection Paradox is not explicitly named but its underlying logic appears in renewal/stopping-time discussions (SP15).
Modeling Empirical Workloads: heavy-tailed, Pareto, tail behavior	Conceptually Related — Renewal theory and distributional variability appear throughout SP14–SP15; however, explicit heavy-tailed/Pareto workload modeling is not discussed.
Management of Server Farms: provisioning, power management, routing policies	Indirectly Supported — SYSU's MDP/DP framework (SP10–SP12) provides the mathematical foundation for such control problems, though no server farm–specific models are developed.
Scheduling Analysis: FCFS, priority classes, PS, LCFS, FB, SJF, PSJF, SRPT, SOAP, Gittins Index, Whittle Index	Partially Covered / Conceptually Related — FCFS appears as the service discipline for G/G/1 and G/G/m models (SP14, SP15). SYSU also includes optimal ordering/scheduling examples within MDP discussions (SP10–SP11), which are Conceptually Related to queueing scheduling. Advanced queueing scheduling (SRPT, PS, Gittins, SOAP, etc.) does not directly appear.
Datacenter Applications: multiserver scheduling, multi-dimensional resource allocation, c-μ rule, parallel speedup models	Indirectly Supported — MDP, dynamic programming, and stochastic modeling techniques provide the theoretical basis (SP10–SP12). No direct datacenter resource or speedup-function models appear, but the mathematical machinery corresponds closely.

Request 3

SYSU Machine Learning ↔ CMU 10-701 Introduction to Machine Learning

SYSU Course Assignments & Slides (attached files)

- **Assessments**

- **Slides**
  - 1 – Introduction & Overview of Machine Learning
  - 2 – Bayesian Classifiers
  - 3 – Maximum Entropy Models & Logistic Regression
  - 4 – Boosting Methods (AdaBoost, Gradient Boosting)
  - 5 – Markov Chain Monte Carlo (MCMC)
  - 6 – Big Data and Big Models
  - 7 – Pretrained Models and Representation Learning
  - 8 – Prompt-based Learning
  - 9 – Introduction to Causal Inference
  - 10 – Causal Models (I): Backdoor Adjustment
  - 11 – Causal Models (II): Frontdoor Adjustment
  - 12 – Causal Models (III): Counterfactuals
  - 13 – Reinforcement Learning (Part I: Fundamentals and MDPs)
  - 14 – Reinforcement Learning (Part II: Model-Based RL)
  - 15 – Imitation Learning and Policy Optimization

CMU Course Reference Webpage: https://www.cs.cmu.edu/~hchai2/courses/10701/#Syllabus

1. Course Information

Category	**SYSU Course**	**CMU Course**
Course Title	Machine Learning	Introduction to Machine Learning
Institution	Sun Yat-sen University (SYSU), School of Computer Science and Engineering	Carnegie Mellon University (CMU), Machine Learning Department
Course Code	-	10-701
Instructors	• Prof. Yanghui Rao (raoyangh@mail.sysu.edu.cn) • Prof. Hanjiang Lai (laihanj3@mail.sysu.edu.cn) • Prof. Ning Tan (tann5@mail.sysu.edu.cn)	Prof. Henry Chai
Credit Hours	3 credits (54 contact hours)	12 units
Textbooks	• Zhihua Zhou, Machine Learning, Tsinghua University Press, 2016. • Christopher M. Bishop, Pattern Recognition and Machine Learning, Springer, 2006.	No single required textbook.
Assignments	Three module-based assessments (classification (in-class test), causal inference (report), RL (in-class test)); one coding-based presentation For the presentation, students are required to implement and analyze models and present formal reports,	Six homework assignments combining programming and written components (HW1–HW4: decision trees, kNN, regression, probabilistic models, neural networks and deep learning; HW5–HW6: unsupervised learning, reinforcement

	including mathematical derivations and experimental results.	learning, learning theory, ensemble methods). One course project with proposal, mid-term check-in, and final report focusing on applying/advancing ML methods on a real-world or research problem.
Exams / Deliverables	-	Two written exams: • Midterm exam (closed-book with one A4 cheatsheet). • Final exam (closed-book with one A4 cheatsheet). Multiple homework assignments and a term project (proposal, check-in, final report).
Grading	Class Participation 40% Testing / Reports 60%	• Participation (in-class polls): 5% • Midterm exam: 25% • Final exam: 25% • Homework assignments (6): 25% • Course project: 20%
Official Webpage	-	https://www.cs.cmu.edu/~hchai2/courses/10701/#Syllabus

2. Course Description Comparison

Aspect	**SYSU Machine Learning**	**CMU 10-701 Introduction to Machine Learning**
Course Focus	Provides a comprehensive graduate-level foundation in modern machine learning, combining probabilistic models, discriminative and generative frameworks, and recent paradigms including pretrained models, prompt learning, causal inference, and reinforcement learning.	This course is designed to give PhD students a solid foundation in the methods, mathematics, and algorithms of modern machine learning. Covers core algorithms for classification, regression, clustering, representation learning, neural networks/deep learning, unsupervised learning, and reinforcement learning. Emphasizes formal derivations, assumptions, and properties of learning algorithms, as well as experimental design, model selection, and analysis of modern ML systems.
Learning Objectives	1. Understand probabilistic and statistical principles of ML. 2. Apply discriminative & generative models for supervised learning. 3. Implement ensemble and Bayesian inference methods.	By the end of 10-701, students are expected to: 1. Have a deep understanding of the assumptions, derivations, and usage of key machine learning algorithms. 2. Implement and analyze learning algorithms for classification, regression, structured prediction, clustering, and representation learning.

	4. Explain pretrained and prompt-based paradigms. 5. Understand causal reasoning (backdoor/frontdoor/counterfactual). 6. Apply MDPs and RL algorithms for decision-making.	3. Integrate multiple components of practical ML systems, including data preprocessing, learning, regularization, and model selection. 4. Describe and derive formal properties of models and algorithms, and interpret their practical implications. 5. Compare and contrast different learning paradigms (supervised, unsupervised, reinforcement learning). 6. Design experiments to evaluate and compare learning methods on real-world problems. 7. Employ probability, statistics, calculus, linear algebra, and optimization to develop new predictive models or learning methods. 8. Analyze and synthesize academic literature in a current research area of machine learning and propose potential future directions.

3. Topic Mapping and Content Equivalence

CMU Topic	**SYSU Correspondence**
Introduction: Notation & Problem Formulation	Partially Covered – SYSU introduces ML definitions, paradigms, and problem setup, though notation systems differ. (PPT 1)
Decision Trees	Conceptually Related – Boosting slides mention weak learners conceptually linked to trees, but decision tree algorithms are not developed. (PPT 4)
KNNs & Model Selection	Conceptually Related – Introduction mentions classical ML models broadly; no direct treatment of kNN or model selection theory. (PPT 1)
Linear Regression	Indirectly Supported – Linear models appear as special cases within exponential family/log-linear modeling, but no dedicated linear regression lecture. (PPT 3)
MLE / MAP	Partially Covered – Bayesian classifier uses MAP decision rules; Maximum Entropy model uses constrained likelihood maximization. (PPT 2, PPT 3)
Naïve Bayes	Covered – SYSU provides classical Bayesian generative classifier formulation including conditional independence assumptions. (PPT 2)

Logistic Regression	Covered – Maximum Entropy model connects directly to logistic regression through log-linear models and softmax/ logistic forms. (PPT 3)
Regularization	Conceptually Related – Appears implicitly via constraints in MaxEnt and complexity control in Boosting; not a standalone topic. (PPT 3, PPT 4)
Neural Networks	Conceptually Related – SYSU references neural networks only at high level in big-model discussions; no foundational NN lecture. (PPT 6)
Backpropagation	Indirectly Supported – Mentioned as training mechanism for deep models, but no derivation or dedicated treatment. (PPT 6)
CNNs	Indirectly Supported – CNNs appear only as representative vision backbones in big-model slides; operations are not explained. (PPT 6)
RNNs	Indirectly Supported – RNN/LSTM/GRU appear in sequence-model context but without mathematical development. (PPT 6)
Attention & Transformers	Partially Covered – SYSU covers BERT/GPT, attention-based pretraining, and prompt-tuning, but not transformer mathematics. (PPT 6, PPT 7)
Unsupervised Learning: Clustering	Conceptually Related – Some unsupervised/self-supervised representation learning appears, but classical clustering is absent. (PPT 6)
Unsupervised Learning: Dimensionality Reduction	Conceptually Related – Encoder-based representations appear in self-supervised contexts, but no PCA/linear DR focus. (PPT 6)
Reinforcement Learning: Value & Policy Iteration	Covered – SYSU gives complete MDP formulation, Markov chains, Bellman expectation/optimality equations, and dynamic programming (VI/PI). (PPT 12, PPT 13)
Reinforcement Learning: Q-Learning & Deep RL	Partially Covered – SYSU covers Q-functions, Bellman optimality, and families of Deep RL methods (DQN, PG, PPO, SAC) at an overview level. (PPT 12, PPT 13)
Pretraining, Fine-tuning & In-context Learning	Covered – SYSU gives detailed treatment of self-supervised pretraining, large models, fine-tuning, and prompt-based / in-context learning. (PPT 6, PPT 7)
Algorithmic Bias	Conceptually Related – SYSU's causal inference modules provide tools (confounding, adjustment, counterfactuals) relevant to fairness analysis but not framed as algorithmic bias. (PPT 8, PPT 9, PPT 10, PPT 11)

Learning Theory I	Conceptually Related – Some learning-theoretic ideas appear implicitly in MaxEnt and RL convergence, but no general-purpose statistical learning theory module. (PPT 3, PPT 13)
Learning Theory II	Conceptually Related – As above, SYSU lacks VC/PAC-style generalization theory; only isolated conceptual overlap. (PPT 3, PPT 13)
Ensemble Methods: Bagging	Indirectly Supported – SYSU's ensemble discussion is Boosting-focused; bagging appears only as a general ensemble concept. (PPT 4)
Ensemble Methods: Boosting	Covered – SYSU devotes an entire lecture to AdaBoost and gradient boosting, matching CMU's treatment closely. (PPT 4)
SVMs	Conceptually Related – Margin-based ideas appear indirectly in inverse RL (max-margin IRL), though classical SVM theory is absent. (PPT 15)
Kernels	Indirectly Supported – Gaussian Processes in model-based RL reference kernelized covariance structures, but kernel methods are not taught systematically. (PPT 14)
Gaussian Processes	Partially Covered – SYSU uses Gaussian Processes for model-based RL dynamics modeling; however, GP regression theory is not the focus. (PPT 14)
Generative Models for Vision	Conceptually Related – SYSU discusses self-supervised vision encoders (MAE, DINO, BEiT), but not generative models like GANs or diffusion. (PPT 6)

Request 4

SYSU Pattern Recognition (PR) & Computational Visual Media (CVM) ↔
CMU 11-777 Multimodal Machine Learning

Pattern Recognition and Computational Visual Media were designed as a consecutive sequence forming a single graduate track on multimodal learning, with Pattern Recognition providing the multimodal foundation and CVM focusing on visual-language integration and multimodal generation. Both of them provide comprehensive coverage of multimodal learning, integrating vision, language, and sequence modeling. I therefore request that these two courses be jointly recognized as equivalent to CMU 11-777.

SYSU Pattern Recognition (PR) & Computational Visual Media (CVM)

- **Assessments**
- **Slides**
  - PR01 – Overview of Artificial Intelligence
  - PR02 – Bayesian Foundations and Classification

- PR03 – Linear and Nonlinear Classifiers
- PR04 – Artificial Neural Networks
- PR05 – HMM and Markov Models
- PR06 – Deep Sequence Models
- PR07 – Ensemble Learning and Network Optimization
- PR08 – Unsupervised Learning
- PR09 – Generalized Weakly Supervised Learning
- PR10 – Dimensionality Reduction & Model Compression
- PR11 – Basics of NLP
- PR12 – Computer Vision: Image Feature Extraction & Understanding
- PR13 – Computer Vision: Video Analysis – Feature Extraction
- PR14 – Computer Vision: Video Understanding & Vision–Language
- PR15 – Data Mining for Recommender Systems
- CVM01 – Geometry-Driven Image Processing
- CVM02 – Filtering-Based Computational Imaging (Part 1)
- CVM03 – Filtering-Based Computational Imaging (Part 2)
- CVM04 – Deep Learning in Approximation Viewpoint
- CVM05 – Video Processing Fundamentals
- CVM06 – Video Text Information Processing
- CVM07 – Educational Video Processing
- CVM08 – Video Summarization and Thumbnail Generation
- CVM09 – Introduction to Sketch
- CVM10 – Sketch Data Construction
- CVM11 – Sketch Understanding
- CVM12 – Sketch Colorization

CMU Course Reference Webpage: https://cmu-multicomp-lab.github.io/mmml-course/fall2023/

1. Course Information

Category	**SYSU Courses**	**CMU Course**
Course Title	Pattern Recognition; Computational Visual Media	Multimodal Machine Learning
Institution	Sun Yat-sen University (SYSU), School of Computer Science and Engineering	Carnegie Mellon University (CMU), Language Technologies Institute
Course Code	-	11-777
Instructors	1. PR: Prof. Zheng Weishi (Coordinator), Prof. Wang Ruixuan, Prof. Yang Meng, Prof. Wang Changdong, Prof. Hu Jianfang, Prof. Xie Xiaohua, Prof. Ma Jinhua; 2. CVM: Prof. Zhuo Su, Prof. Ruomei Wang, Prof. Chengying Gao	Prof. Louis-Philippe Morency, Co-lecturer: Paul Liang
Credit Hours	3 credits each (54 contact hours each; total 6 credits, 108 hours)	12 units

Textbooks	No single textbook; references include recent papers from CVPR/ICCV/ECCV, ACM TOG, IEEE TPAMI, and AI/CV textbooks (e.g., on Bayesian decision theory, deep learning)	Reading material will be based on published technical papers available via the ACM/IEEE/Springer digital libraries or freely available online (e.g., arxiv.org).
Assignments	PR: Survey report, group project report/experiments/presentation, mid-term progress check; CVM: Three survey reports with code/figures/references	▪ Dataset and project preferences– Let us know your preferences for the datasets that you would like to use for the course project. This will help with the team matching process. ▪ Project Pre-proposal– You should have selected your teammates, dataset, and task. Submit a 1-page pre-proposal plan. ▪ First assignment.This assignment requires a good literature review on the topic of your proposed project. ▪ Second assignment– This assignment focuses on unimodal representations and dataset analyses. ▪ Midterm assignment – Students are asked to implement and evaluate state-of-the-art baseline models on their project dataset and perform error analysis. ▪ Final assignment – Students should explore new ideas to model their multimodal research project
Exams	-	-
Grading	PR: Individual survey report (40%), group CVPR-style English project paper (50%), participation (10%); CVM: Three survey reports (30% each), class participation (10%)	Lecture participation and highlights 16% Reading assignments 12% Course project assignments ▪ Project preferences and pre-proposal 2% ▪ First project assignments 10% ▪ Second project assignment 10% ▪ Mid-term report and presentation 20% ▪ Final report and presentation 30%
Official Webpage	PR: https://csai-sysu.net/aicourse/HTML/home.html; CVM: -	1. https://cmu-multicomp-lab.github.io/mmml-course/fall2023/ 2. https://cmu-multicomp-lab.github.io/mmml-course/assets/pdf/MultimodalML-Fall2023-Syllabus.pdf

2. Course Description Comparison

Aspect	**SYSU Pattern Recognition & Computational Visual Media**	**CMU 11-777 Multimodal Machine Learning**

Course Focus	**PR**: Foundations of statistical pattern recognition (Bayesian theory, classifiers, NNs, HMMs, deep models), representation/compression (dim reduction, model compression), applications in NLP, image/video understanding, recommenders. **CVM**: Geometry/filtering-based imaging, video understanding/visualization (structure analysis, summarization, retrieval), sketch computing (data, understanding, applications). Combined: Emphasizes vision-language integration, multimodal video analysis, temporal modeling, and heterogeneous representations for perception tasks.	The course will present the fundamental mathematical concepts in machine learning and deep learning relevant to the six main challenges in multimodal machine learning: (1) representation, (2) alignment, (3) reasoning, (4) generation, (5) transference and (6) quantification. These include, but not limited to, multimodal transformers, neuro-symbolic models, multimodal tensor fusion, mutual information and multimodal graph networks. The course will also discuss many of the recent applications of MMML including multimodal affect recognition, multimodal language grounding and language-vision navigation.
Learning Objectives	**PR**: Explain Bayesian frameworks/classifiers, implement generative/discriminative models/NNs/sequence models, apply ensemble/unsupervised/weakly supervised methods, use dim reduction/compression, understand NLP/CV/video pipelines, conduct surveys/projects. **CVM**: Explain edge-aware filtering/optimization links, compare video/sketch approaches, analyze papers for surveys connecting to research. Combined: Build skills in multimodal perception (vision-text-video), feature extraction/alignment, generation/retrieval, and critical analysis for AI applications.	-

3. Topic Mapping and Content Equivalence

CMU Lecture	**SYSU Coverage**
Week 1 – Introduction, MMML Challenges, Tasks & Datasets	Conceptually Related – SYSU provides AI background, modality pipelines, and course framing relevant to MMML context. (PR01, CVM01)

Week 2 – Unimodal Representations (vision, language, audio, graphs)	Partially Aligned – Strong visual and sequence representations; NLP and HMM/Transformer support; audio minimal, graph mainly in sketch. (CVM01–03, CVM04, PR11, PR12, PR13, PR05, PR06)
Week 3 – Multimodal Representations (fusion, coordinated, hybrid)	Partially Aligned – SYSU supports multimodal video–text–audio fusion at task level but not general MM representation frameworks. (CVM05–08, PR14)
Week 4 – Alignment & Grounding (attention, alignment, self-supervision)	Partially Aligned – Video–text/slide alignment and complete attention/Transformer machinery provide practical and architectural support. (CVM06-07, PR06)
Week 5 – Multimodal Transformers & Reasoning	Conceptually Related – Full Transformer architecture and multigraph transformers support structured modeling; reasoning introduce. (PR06, CVM11)
Week 6 – Multimodal Language Grounding (captioning, vision–language)	Partially Aligned – Strong grounding in video–text, slide–speech, and vision–language captioning tasks. (CVM06-08, PR14)
Week 7 – Multimodal Interaction & Inference (RL, causal, symbolic)	Conceptually Related – SYSU offers causal inference and structured reasoning foundations. (PR02, PR04, CVM11)
Week 9 – Multimodal Generation (summarization, conditional generation)	Strongly Aligned – Extensive conditional generation (automatic, user-guided, text-guided, reference-guided, few-shot) and multimodal video summarization. (CVM12, CVM08, PR04)
Week 10 – Midterm Project Presentations	Strongly Aligned – SYSU PR includes research-style projects, mid-term reviews, and structured presentation cycles. (PR Project Weeks 15–18)
Week 11 – Transference (modality transfer, co-learning, self-training)	Partially Aligned – Weakly supervised learning, co-training, semi-supervision, few-shot and domain adaptation tasks provide methodological alignment. (PR09, PR Project Topics, CVM12)
Week 12 – Quantification (heterogeneity, bias, fairness) & New Directions	Conceptually Related – SYSU includes risk quantification, fairness, interpretability, and emerging-trend discussions. (PR02, PR04)
Week 14 – Guest Lectures	Indirectly Supported – No guest-lecture module, but extensive use of recent CVPR/TPAMI papers and survey assignments plays a similar role. (CVM Reading-Based Slides, PR Project Literature)

Week 15 – Final Project Presentations	Strongly Aligned – SYSU PR ends with full project presentations, Q&A, and evaluation, matching CMU's final presentation structure. (PR Project Weeks 15–18)

Request 5

SYSU Advanced Computer Networks ↔ CMU 15641 Networking and the Internet

SYSU Advanced Computer Networks

- **Assessments**
- **Slides**
  - PPT1-Module 1&2
  - PPT2-Module 3
  - PPT3-Module 4
  - PPT4-Module 5

CMU Course Reference Webpage: https://psteenkiste.github.io/nets-fa24/

1. Course Information

Category	**SYSU Course**	**CMU Course**
Course Title	Advanced Computer Networks	Networking and the Internet
Institution	Sun Yat-sen University (SYSU), School of Computer Science and Engineering	Carnegie Mellon University (CMU), Computer Science Department
Course Code	–	15-641
Instructors	Prof. Xiaoxi Zhang (zhangxx89@mail.sysu.edu.cn)	Prof. Peter Steenkiste
Credit Hours / Units	3 credits (54 contact hours)	12 units
Textbooks	James F. Kurose，Keith W. Ross，"Computer Networking，A Top-Down Approach"，the 7th edition，2017	Recommended: Computer Networks: A Systems Approach, by Larry Peterson and Bruce Davie

Assignments	Class attendance & quizzes; mid-term presentation; final group presentation + report	Homework sets; midterm/final exam; programming projects
Exams	No formal exam; assessed by mid-term + final project	2 exams (midterm + final)
Grading	Attendance/Quiz 20%; Mid-term 10%; Final Presentation & Report 70%	Homework 2.4% * 5, exams (mid 18%, final 25%), and programming projects (3% + 6% * 6 + 2% * 3)
Official Webpage	–	https://psteenkiste.github.io/nets-fa24/

2. Course Description Comparison

Aspect	**SYSU Advanced Computer Networks**	**CMU 15-641 Networking and the Internet**
Course Focus	Graduate-level introduction to modern computer networking: Internet architecture, packet switching, performance modeling, multimedia networking & QoS, network security, SDN/OpenFlow/P4, NFV/middleboxes, wireless/5G systems, IPv6/SRv6. Focus on emerging Internet architectures, programmable networking, wireless/mobile systems, and security protocols used in real deployments.	15-441/641 is an introductory course in computer networks. The emphasis will be on the basic performance and engineering tradeoffs in the design and implementation of computer networks. To make the issues more concrete, the class includes several multi-week projects requiring significant design and implementation.
Learning Objectives	Analytical understanding of QoS, multimedia, cryptographic protocols (TLS/IPsec), SDN/P4, and wireless systems. Research-style final project. Ability to analyze packet-level performance, evaluate QoS, understand secure framework, program SDN/P4, reason about wireless/5G systems, and present research.	The goal is for students to learn not only what computer networks are and how they work today, but also why they are designed the way they are and how they are likely to evolve in the future. We will draw examples primarily from the Internet. Topics to be covered include: congestion/flow/error control, routing, addressing, naming, multi-casting, switching, internetworking, and network security. Evaluation is based on homework assignments, the projects, one midterm exam, and one final exam.

3. Topic Mapping and Content Equivalence

CMU Lecture	SYSU Coverage
Network Transmission & Performance	Fully Covered — Packet switching, queuing, transmission delay, propagation, end-to-end delay directly match Module 1 discussions. Slides: M1
Link Layer	Partially Covered — Layering, framing, MAC addresses, error detection appear in Module 1, though less detailed than CMU. Slides: M1
Ethernet & Switching	Partially Covered — MAC learning switches appear in Wi-Fi section ("switch updates table based on source address"). Ethernet details lighter. Slides: M5
Distance Vector Routing	Conceptually Related — Routing basics appear, but DV algorithm not explicitly derived; LS routing dominates. Slides: M4
Link-State Routing & SDN	Largely Covered — LS routing (Dijkstra), centralized control, match-action forwarding, OpenFlow-style tables all present. Slides: M4
Internet Protocol & Layering	Fully Covered — Detailed IPv4 packet handling, fragmentation example, complete layering model. Slides: M4; M1
Interdomain Routing (BGP)	Largely Covered — eBGP/iBGP, path attributes, policy routing, plus modern extensions (capacity/performance-aware). Slides: M4
Transport Layer Overview	Partially Covered — TCP/UDP roles listed in stack overview; deeper mechanics not expanded. Slides: M1
TCP Windowing & Congestion Avoidance	Indirectly Supported — Only referenced in wireless mobility context ("TCP mistakes losses for congestion"), without algorithmic treatment. Slides: M5
Advanced Congestion Control	Conceptually Related — TE systems (e.g., B4) provide high-level congestion-avoidance motivation but not host-level algorithms. Slides: M4
QoS, Rate Limiting & Performance	Largely Covered — QoS from multimedia angle: jitter, playout buffer, rate adaptation (DASH), streaming constraints. Slides: M2
DNS	Largely Covered — Full CDN lookup workflow includes authoritative DNS, CNAME, resolution sequence. Slides: M2

HTTP & Web Performance	Partially Covered — Streaming over HTTP, DASH client logic, CDN delivery over HTTP. Less about HTTP internals. Slides: M2
Content Distribution	Fully Covered — CDN architecture, deep vs bring-home, KingCDN example, OTT challenges, Netflix case study. Slides: M2
Video on Demand	Fully Covered — Playout buffer, jitter control, DASH chunk selection, adaptive bitrate. Slides: M2
Datacenter Networks	Conceptually Related — SYSU focuses on inter-datacenter TE (Google B4) rather than intra-DC topologies. Slides: M4
Security: Cryptography Basics	Fully Covered — Symmetric/public-key crypto, MAC, signatures, authentication protocols. Slides: M3
Security: TLS/SSL	Fully Covered — Full handshake sequence, key derivation, record structure. Slides: M3
Security: IPsec	Fully Covered — AH/ESP, transport vs tunnel mode, SAs, SPD/SAD, VPN operation. Slides: M3
Routing Security	Conceptually Related — BGP scalability/robustness problems discussed; no explicit RPKI/BGPsec. Slides: M4
Wireless PHY & Wi-Fi	Fully Covered — Path loss, interference, SNR–BER, hidden/exposed terminals, CSMA/CA, 802.11 frame format. Slides: M5
Cellular Networks	Fully Covered — LTE/5G architecture, control/user-plane separation, GTP tunneling, mobility procedures. Slides: M5
Network Functions (NAT, Firewall, NFV)	Largely Covered — Middleboxes, firewalls/IDS, NFV, service chaining integrated into SDN and security modules. Slides: M4, M3